C PROGRAMMING

Introduction & Environment Setup

Lecture 01



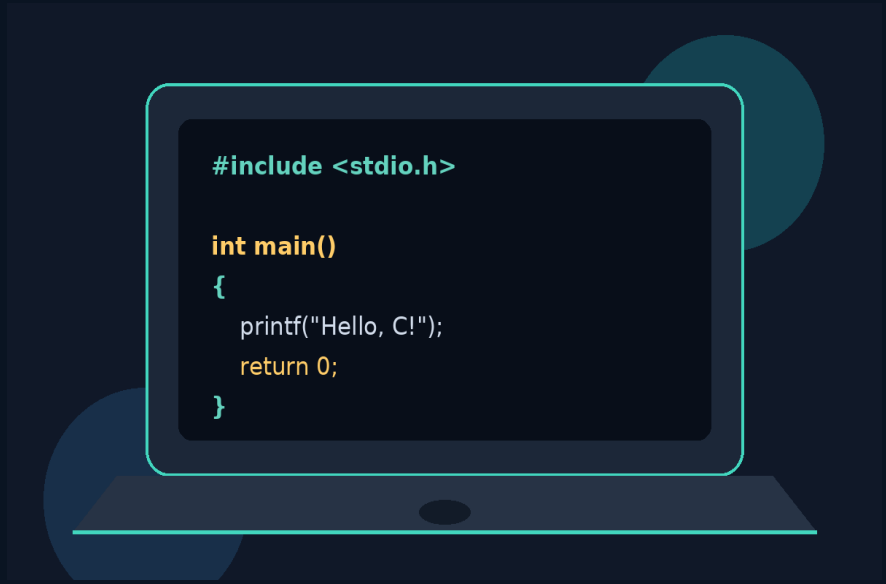
Instructor **Simanto Chandra Kor**

Batch • CSE-60

What is Programming?

Before learning C, let's understand what programming actually means.

- Programming means giving clear instructions to a computer.
- A program is a sequence of instructions that solves a problem.
- Programming languages provide a structured way to write those instructions.
- Example: take two numbers → add them → display the result.



What is C?

A small language with a huge influence.

- General-purpose programming language.
- Created by Dennis Ritchie at Bell Labs.
- Fast, efficient and close to the hardware.
- Used in operating systems, embedded systems, compilers and more.
- A foundation for understanding how computers execute programs.



C

Why Learn C?

The goal is not just to memorize C syntax.

01 **Programming basics**

Variables • conditions • loops • functions

02 **Memory**

Understand how data lives in memory

03 **Problem solving**

Turn ideas into algorithms

04 **Strong foundation**

Useful for C++, DSA & competitive programming

How Does a C Program Run?

From source code to output.



What Do We Need?

Three ideas are enough for your first day.

01

Code Editor

Where we write the source code.

02

Compiler

Translates C code into executable instructions.

03

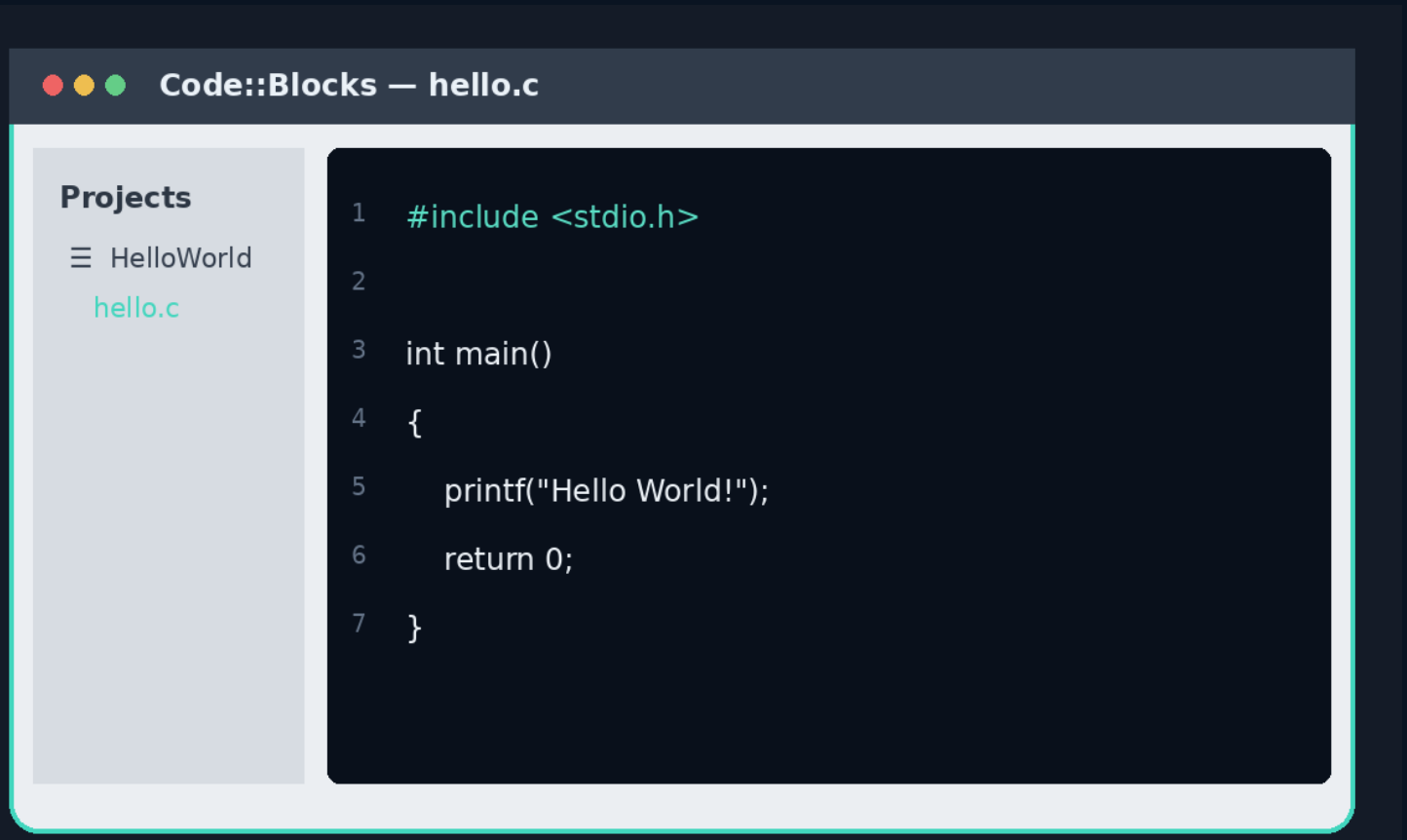
Console

Where the program runs and output appears.

IDE = Integrated Development Environment

Why Code::Blocks?

A beginner-friendly IDE for getting started with C.



- Code editor
- Compiler integration
- Build & Run tools
- Debugger
- Project management

Code::Blocks Setup

For Windows beginners, use the package that includes a compiler.

1 **Download Code::Blocks**

2 **Install**

3 **Open IDE**

4 **New Console Application**

5 **Select C**

6 **Create source file**

Tip: the official Code::Blocks site provides binary releases; the full package bundles a MinGW compiler.

Your First C Program 🎉

Don't worry if every line is not clear yet — we will learn them one by one.

```c
#include <stdio.h>

int main()
{
    printf("Hello World!");

    return 0;
}
```

OUTPUT

Hello World!

Run → Build & Run

Your first executable program.

Understanding the First Program

Only the essential ideas for Lecture 1.

`#include <stdio.h>`	Provides standard input/output functions.
`main()`	The starting point of program execution.
`{ }`	Defines a block of code.
`printf()`	Displays output on the screen.
`return 0;`	Ends the program successfully.

Where Do We Practice?

Learning programming requires practice, not only lectures.



Codeforces

A popular platform for competitive programming and problem solving.

- Solve programming problems.
- Participate in contests.
- Practice by difficulty and rating.
- Read editorials and learn from other solutions.
- For beginners: focus on solving, not rating.



CodeChef

Practice coding, learn concepts, and participate in contests.

- Programming practice problems.
- Contests and competitive programming.
- Beginner-friendly learning content.
- Problems across programming and DSA.
- Use it as an extra practice platform.

CC

CODECHEF

Learn • Practice • Compete

VJudge

One interface for problems from many online judges.

- Aggregates problems from multiple judges.
- Useful for mixed problem sets.
- Can include sources such as Codeforces, CodeChef, AtCoder, UVA and others.
- Especially useful when an instructor gives a curated set.

VJ

VJUDGE

Many judges → One platform

Codeforces vs CodeChef vs VJudge

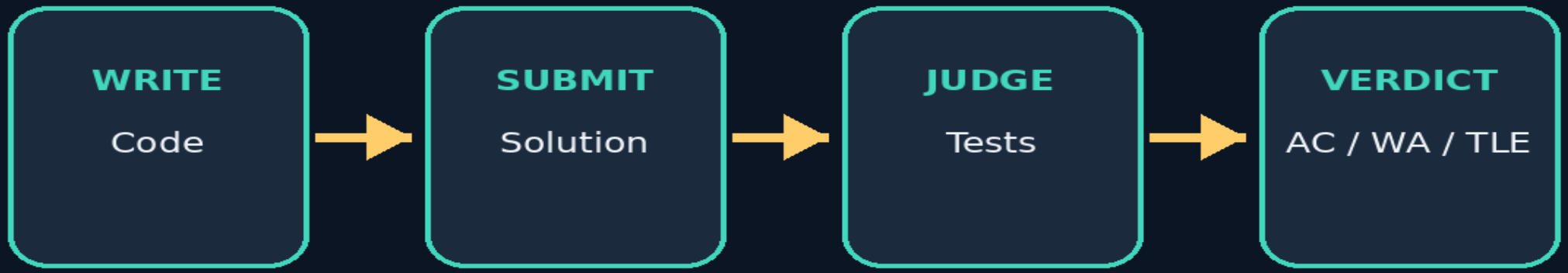
You only need to be familiar with them for now.

Platform	Main idea	Good for
Codeforces	Practice + contests	Competitive programming
CodeChef	Practice + contests	Learning + practice
VJudge	Multiple judges	Mixed problem sets

Don't try to master all three on day one.

How Online Judging Works

Your code is tested automatically against test cases.



Common Online Judge Results

You will see these abbreviations very often.

AC — **Accepted**

Solution is correct.

WA — **Wrong Answer**

Output is incorrect.

TLE — **Time Limit Exceeded**

Program takes too long.

CE — **Compilation Error**

Code could not be compiled.

RE — **Runtime Error**

Program crashes or performs an invalid operation.

Our Learning Journey

We will build skills step by step.

01

C Basics

02

Variables & Data Types

03

Input / Output

04

Conditions(if/else)

05

Loops

06

Arrays & Strings

07

Problem Solving

Goal: understand → practice → solve → improve

First Class Challenge

Write, compile, run — and celebrate your first program.

PRINT THIS:

```
Hello, World!
My name is ______.
I am learning C programming.
```

1. Write
2. Compile
3. Run
4. Submit

FINAL THOUGHT

Programming is learned by doing.

Type it. Break it. Debug it. Solve it. Improve it.

TYPE

Don't just copy

TRY

Experiment

DEBUG

Learn from errors

SOLVE

Practice problems

See you in the next lecture!